

Unit 1

Introduction to linear models.

Unit Goals

- Understand what is meant by a linear equation and a linear model or system.
- Understand what is meant by a mathematical model.
- Be able to formulate a mathematical model for a problem, given a description of the problem in plain English.
- Begin to recognize models of several common problem classes.
- Understand that finding solutions, where they exist, is often non-trivial.

Unit 1.1

What is Linear Algebra and why should you care?

Why study linear algebra?

- Most of us study any branch of math for one primary reason: to enable us to solve problems that have a mathematical basis.
- Mathematical models are central to problems in all branches of the sciences and engineering, most of the social sciences, and a growing subset of the arts.
- Most problems of interest are multi-dimensional; they involve many different variables.
- **Linear Algebra and Calculus III are the two courses that teach the fundamental concepts, language and tools necessary to work with multi-variable mathematical models.**

Why study linear algebra?

Linear algebra plays a central role in many applications across many disciplines:

- Statistics/Data Analysis (Principal Component Analysis; Regression)
- Industrial Engineering/Operations Research (Linear Programming)
- Mechanical Engineering & Physics
- Electronic Engineering (Very Large-Scale Integration, VLSI)
- Chemical Engineering & Computational Biology
- Machine Learning (Dimensionality Reduction, Deep Learning)
- Natural Language Processing
- Image Processing & Computer Graphics

Why study linear algebra?

- Linear algebra provides an **extension to the language of math** that greatly simplifies how we write systems of many equations in many variables.
- Linear algebra provides **a set of tools and methods** that allows entire classes of problem to be solved in a straightforward and robust method.
- Linear algebra specifically provides a framework for working with **linear mathematical models** but is central to how many non-linear models are solved as well.

What is a mathematical model?

- A mathematical model is simply a mathematical formulation of a “real world” problem.
- Mathematical models often take the form of one or more equations that describe the problem.
- Classes of related problems can be described with a common model formulation.
- Methods of solving/using mathematical models vary with model type and the exact question being asked.

Mathematical modeling is central to many fields.

- Climate Modeling and Prediction
- Epidemiology
- Computational Pharmacology
- Vehicle Design (Aircraft, Electric vehicles, etc)
- Neuroscience
- Massive Data Analysis
- Risk Assessment and Mitigation
- Economics & Impacts of Public Policy
- Next-Generation Energy Solutions (Battery design, Smart electric grids, etc).

What is a linear model?

- Most simply, a linear model is one in which all the equations contain only LINEAR terms.
- A linear term is one of the general form ax , where a is a constant (numeric or symbolic) and x is a variable.
- For example, the following are all linear terms:

$$3x \quad 2y \quad bt \quad 105z \quad (4ac)x$$

- While these are non-linear:

$$\sin(x) \quad x^2 \quad xy \quad x \ln x \quad e^{-x}$$

Variables vs unspecified constants.

A **variable** is a symbolic representation of a value that may take on different values within the context of a mathematical model:

- There may be a single “correct” value (i.e. the solution), but alternate values have well-defined meanings that do not affect the underlying model.

A **constant** is a value that remains fixed within the context of a single mathematical model:

- Some constants are represented simply by numbers, but others by symbols that represent the constant.
- With unspecified constants, we may not know what the value is, but we know it must be fixed; changing the constant would change an important detail of the model.

Variables vs unspecified constants.

A note on conventions:

- Letters toward the end of the alphabet are generally used for variables, e.g. x, y, z, t ;
- Letters towards the beginning of the alphabet are generally used for constants, e.g. a, b, c, d .
- This convention on the choice of symbols for constants and variables is not always followed by everyone, but will be maintained in this class, unless otherwise specified:
 - In an expression like bt , assume b is a constant and t is a variable.
 - In an expression like $(4ac)x$, assume a and c are both constants and x is a variable.

What is a linear model?

- A linear equation is an equation with **ONLY** linear terms.

- The general form of a linear equation is:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

- Some examples of linear equations:

$$y = 2x \quad z = ax - by \quad x + y = 5 \quad 2x_1 + x_2 + 3x_3 = 7$$

- Some examples of non-linear equations:

$$y = x^2 \quad z = \sin(x + y) \quad 2x + xy + 3y = 7$$

- **If any term in an equation is non-linear, the equation is non-linear.**

What is a linear model?

- A system of equations is a set of equations that use the same variables:
 - *The variables have the same meaning, and must take on the same value, in each equation in the system.*
- A linear system of equations is simply a system in which every equation is linear.
- Some examples of linear systems of equations:

$$\begin{array}{rcl} x & + & y = 1 \\ x & - & 2y = 3 \end{array} \quad \begin{array}{rcl} x & + & y + z = 0 \\ 2x & - & y - 2z = -1 \\ & & y - z = 2 \end{array} \quad \begin{array}{rcl} ax_1 & + & bx_2 = 0 \\ cx_1 & + & dx_2 = 0 \end{array}$$

- If any equation in a system is non-linear, the system is non-linear.
- **A linear model is a mathematical model described by a linear system of equations.**

The geometry of linear systems.

- The general form of a linear equation in 2-D is:

$$a_1x_1 + a_2x_2 = b \rightarrow x_2 = -\frac{a_1}{a_2}x_1 + \frac{b}{a_2}, a_2 \neq 0$$

- This is of the form $y = mx + b$, the equation for a straight line with slope m and y-intercept b .
- A system of linear equations in 2-D is thus a set of (straight) lines.
- Solutions to the system are points where the lines intersect.

The geometry of linear systems.

- A system of two lines in 2-D.

$$2x - 3y = 3 \rightarrow y = \frac{2}{3}x - 1$$

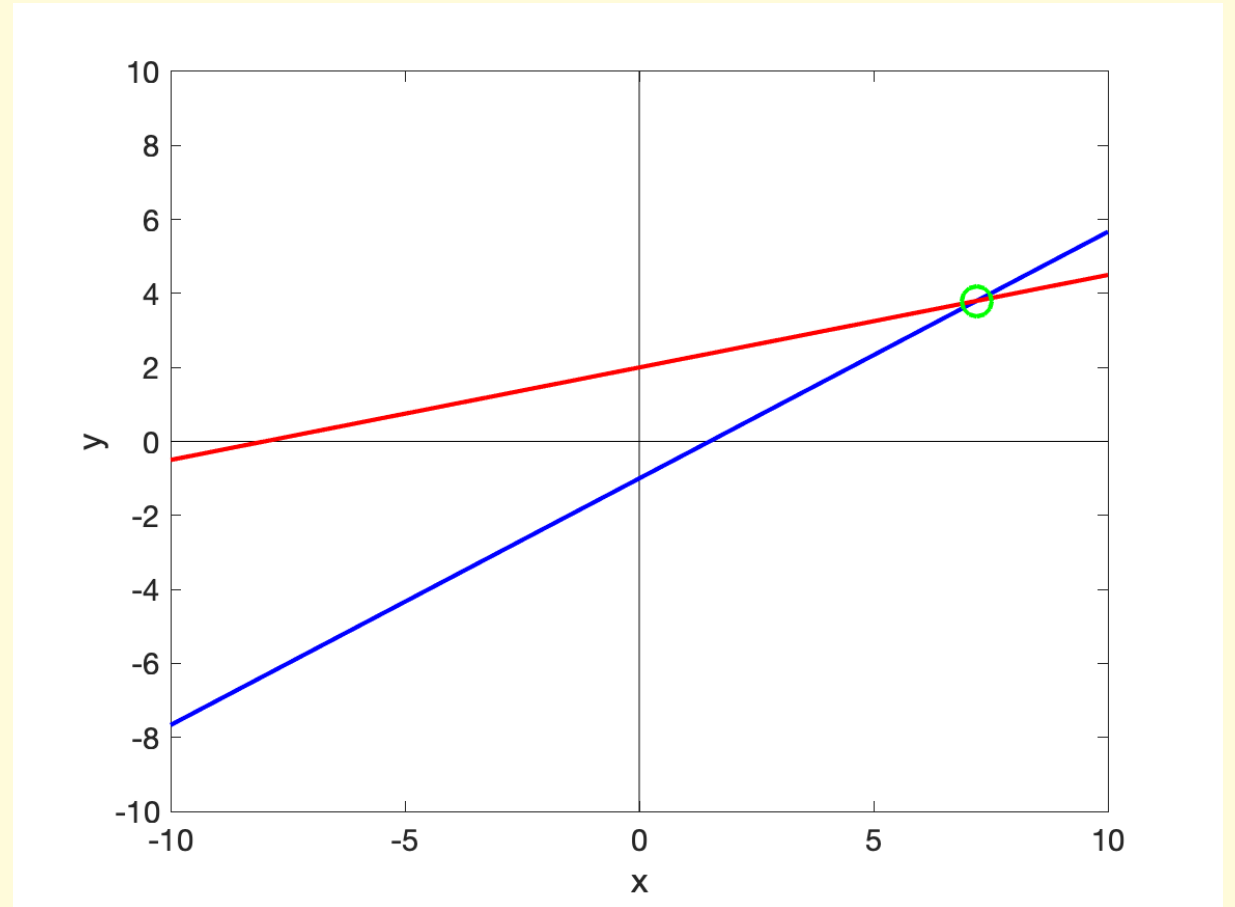
$$x - 4y = -8 \rightarrow y = \frac{1}{4}x + 2$$

- Solutions to the system are points where the lines intersect:

$$\frac{2}{3}x - 1 = \frac{1}{4}x + 2 \rightarrow \frac{5}{12}x = 3 \rightarrow x = \frac{36}{5}$$

$$y = \frac{1}{4}x + 2 = \frac{36}{20} + 2 = \frac{19}{5}$$

Solution: $(x, y) = (7.2, 3.8)$



The geometry of linear systems.

- The general form of a linear equation in 3-D is:

$$a_1x_1 + a_2x_2 + a_3x_3 = b$$

- This is of the form $ax + by + cz = d$, the equation for a plane with normal vector (a, b, c) [and z-axis intercept of $(0, 0, d/c)$].
- A system of linear equations in 3-D is thus a set of planes.
- Solutions to the system are points where the planes intersect.

The geometry of linear systems.

- A system of two planes in 3-D.

$$x - y + z = 4$$

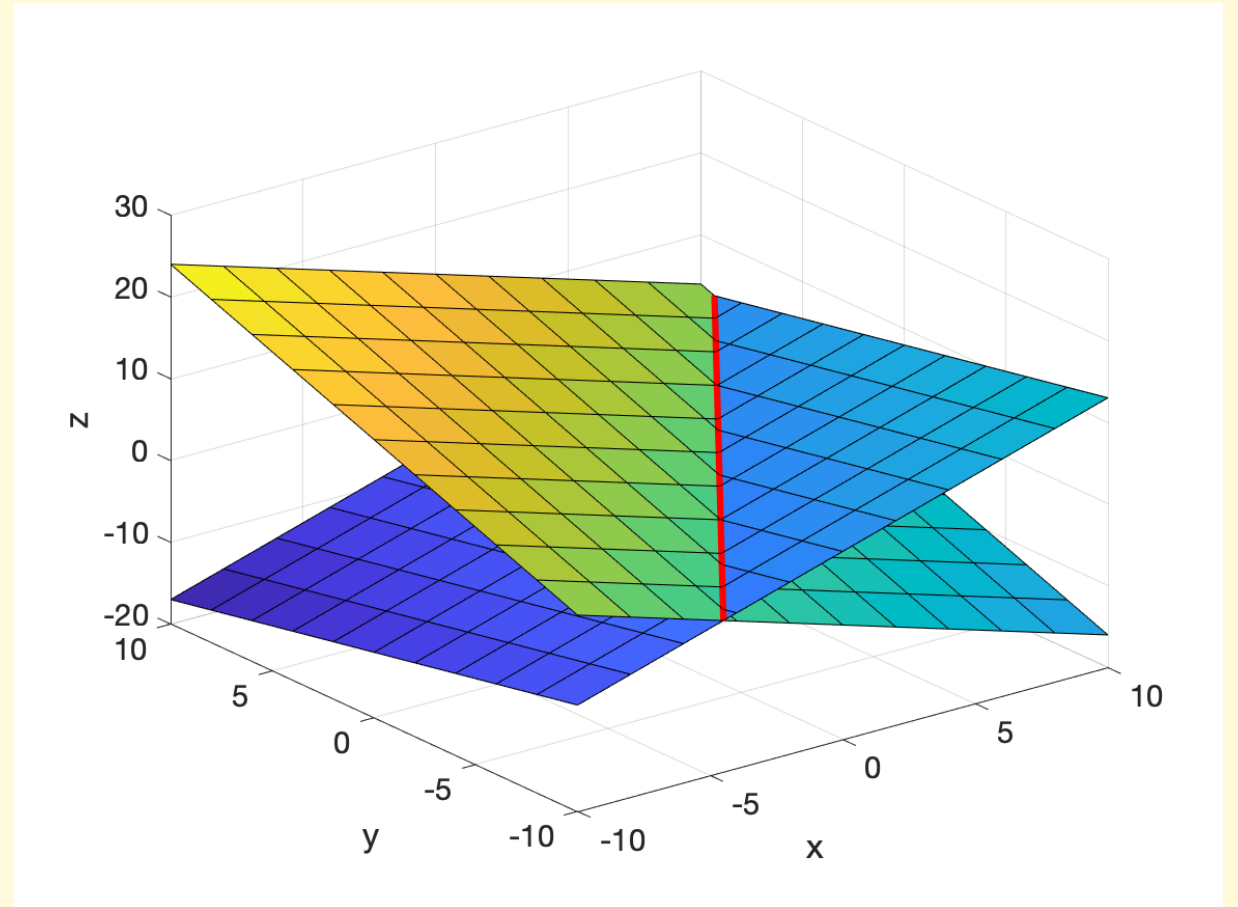
$$x - \frac{y}{2} - z = 2$$

- Solutions to the system are points where the planes intersect:

$$2x - \frac{3y}{2} = 6 \rightarrow x = \frac{3y}{4} + 3$$

$$-\frac{y}{2} + 2z = 2 \rightarrow z = \frac{y}{4} + 1$$

Solution: $(x, y, z) = \left(\frac{3t}{4} + 3, t, \frac{t}{4} + 1\right), \forall t$



The geometry of linear equations.

- A linear equation in 2-D is a line; a line is a 1-dimensional “flat” object.
- A linear equation in 3-D is a plane; a plane is a 2-dimensional “flat” object.
- **A linear equation in any number of variables represents a “flat” geometric object (line, plane, etc.) with a dimension exactly one less than the “parent” space.**
 - We call this a *hyperplane* in dimensions greater than 2.
- Solutions to systems of linear equations are points where all the lines, planes, or hyperplanes intersect!
- More on this later.

Unit 1.1.1

**Worked problems in identifying
linear systems.**

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{rclcl} x^2 & + & y^2 & = & 1 \\ x^2 & - & y^2 & = & 1 \end{array}$$

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{rclcl} x^2 & + & y^2 & = & 1 \\ x^2 & - & y^2 & = & 1 \end{array}$$

- This is a non-linear model, due to the presence of squared variables (x^2, y^2).

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{rclcl} 2x & + & y & = & 0 \\ x & - & 2y & = & 0 \end{array}$$

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{rclcl} 2x & + & y & = & 0 \\ x & - & 2y & = & 0 \end{array}$$

- This is a linear model. All terms are either constants or of the general linear form of constant $\times$ variable.

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{rclcl} \sin \theta & + & \cos \phi & = & 1 \\ \sin 2\theta & - & \tan \phi & = & 0 \end{array}$$

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{rclcl} \sin \theta & + & \cos \phi & = & 1 \\ \sin 2\theta & - & \tan \phi & = & 0 \end{array}$$

- This is a non-linear model, due to the presence of trigonometric terms (sin, cos, tan).

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{rclcl} 3\theta & + & \phi & = & 2 \\ \theta & - & 2\phi & = & -1 \end{array}$$

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{rclcl} 3\theta & + & \phi & = & 2 \\ \theta & - & 2\phi & = & -1 \end{array}$$

- **This is a linear model. All terms are either constants or of the general linear form of constant $\times$ variable.**
- NOTE: Greek letters are often used to denote variables (and symbolic constants) in certain classes of model. Nothing changes when different symbols are used.

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$x + y + z = 0$$

$$x + y + z = -1$$

$$x + y + z = 2$$

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{rcccccc} x & + & y & + & z & = & 0 \\ x & + & y & + & z & = & -1 \\ x & + & y & + & z & = & 2 \end{array}$$

- **This is a linear model. All terms are either constants or of the general linear form of constant $\times$ variable.**
- **NOTE:** This system doesn't have an exact solution, since all the left-hand sides are identical, but the right-hand sides are different. The existence or non-existence of solutions does not affect whether the system is linear.

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{rcccccc} 2x & - & y & + & z & = & 3 \\ x & + & xy & - & z & = & 1 \\ x & - & y & + & 3z & = & 1 \end{array}$$

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{rcccccc} 2x & - & y & + & z & = & 3 \\ x & + & xy & - & z & = & 1 \\ x & - & y & + & 3z & = & 1 \end{array}$$

- This is a non-linear model, due to the presence of a product of variables (xy).

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations, and briefly explain your reasoning.

$$ax - y + z = 0$$

$$x + aby - z = 0$$

$$x - y + abc z = 0$$

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$ax - y + z = 0$$

$$x + aby - z = 0$$

$$x - y + abcz = 0$$

- This is a linear model. All terms are either constants or of the general linear form of constant $\times$ variable.
- NOTE: The linearity of this system rests on the assumption that a, b and c are symbolic constants, rather than variables; this is the most natural interpretation of the given system.

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & a_{13}x_3 & + & a_{14}x_4 & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & a_{23}x_3 & + & a_{24}x_4 & = & b_2 \\ a_{31}x_1 & + & a_{32}x_2 & + & a_{33}x_3 & + & a_{34}x_4 & = & b_3 \\ a_{41}x_1 & + & a_{42}x_2 & + & a_{43}x_3 & + & a_{44}x_4 & = & b_4 \end{array}$$

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & a_{13}x_3 & + & a_{14}x_4 & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & a_{23}x_3 & + & a_{24}x_4 & = & b_2 \\ a_{31}x_1 & + & a_{32}x_2 & + & a_{33}x_3 & + & a_{34}x_4 & = & b_3 \\ a_{41}x_1 & + & a_{42}x_2 & + & a_{43}x_3 & + & a_{44}x_4 & = & b_4 \end{array}$$

- **This is a linear model. All terms are either constants or of the general linear form of constant $\times$ variable.**
- **NOTE:** The linearity of this system rests on the assumption that all a_{ij} and b_i are symbolic constants, rather than variables. The notation used here is in fact one of the most commonly forms used to represent a general linear system.

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$2x - y + z = x + y - z$$

$$x + y - z = 2x + y + z$$

$$2x - y + 3z = x - y + z$$

Identification of linear models.

- Classify the following as either linear or non-linear systems of equations and briefly explain your reasoning.

$$\begin{array}{rcccccccl} 2x & - & y & + & z & = & x & + & y & - & z \\ x & + & y & - & z & = & 2x & + & y & + & z \\ 2x & - & y & + & 3z & = & x & - & y & + & z \end{array}$$

- **This is a linear model. All terms are either constants or of the general linear form of constant $\times$ variable.**
- **NOTE:** While this system is not simplified, with variable terms on both sides of the equations, it still satisfies all the requirements for linearity.

Unit 1.2

Formulating Linear Models

Solving problems with mathematical models.

Key steps:

1. Formulate the model:
 - Carefully define the variables in the model.
 - Write one or more equations that capture all the provided detail.
2. Think about how to solve the problem.
 - What should a solution “look” like; what do you want to find?
 - What method is most appropriate to get this solution?
3. Try to solve the problem:
 - Find the mathematical solution.
 - Interpret the mathematical solution in the context of the problem.

Solving problems with mathematical models.

Possible methods for solving a problem:

- Algebraic manipulation.
- Application of calculus.
- Graphical analysis.
- Estimation.
- Iteration.
- Recursion.
- **New methods from Linear Algebra.**

A classic “Word Problem.”

Miguel spends \$7 to buy 6 apples and 6 oranges. Liz buys 2 apples, 3 oranges and 1 mango for \$4. Wei buys 4 apples and 3 mangos, spending \$5. How much do apples, oranges and mangos cost?

A classic “Word Problem.”

Miguel spends \$7 to buy 6 apples and 6 oranges. Liz buys 2 apples, 3 oranges and 1 mango for \$4. Wei buys 4 apples and 3 mangos, spending \$5. How much do apples, oranges and mangos cost?

Formulate the model:

- Define variables, then cast the information given into a system of equations.

Think about solutions

- What are you trying to find? What method(s) can you use?

Execute the solution:

- First solve the mathematical problem, then interpret your answer in terms of the original problem.

A classic “Word Problem.”

Miguel spends \$7 to buy 6 apples and 6 oranges. Liz buys 2 apples, 3 oranges and 1 mango for \$4. Wei buys 4 apples and 3 mangos, spending \$5. How much do apples, oranges and mangos cost?

Variables are the unknown parts of the problem: the cost of each type of fruit:

- Let x be the cost of apples, y be the cost of oranges, and z be the cost of mangoes (each as a cost for one fruit).

A classic “Word Problem.”

Miguel spends \$7 to buy 6 apples and 6 oranges. Liz buys 2 apples, 3 oranges and 1 mango for \$4. Wei buys 4 apples and 3 mangos, spending \$5. How much do apples, oranges and mangos cost?

Let x be the cost of apples, y be the cost of oranges, and z be the cost of mangoes (each as a cost for one fruit).

Each person’s purchase leads to one equation:

- Miguel: $6x + 6y = 7$
- Liz: $2x + 3y + z = 4$
- Wei: $4x + 3z = 5$

A classic “Word Problem.”

Miguel spends \$7 to buy 6 apples and 6 oranges. Liz buys 2 apples, 3 oranges and 1 mango for \$4. Wei buys 4 apples and 3 mangos, spending \$5. How much do apples, oranges and mangos cost?

Let x , y , and z be the cost of apples, oranges and mangoes, respectively.

$$6x + 6y = 7$$

$$2x + 3y + z = 4$$

$$4x + 3z = 5$$

We want to find the values of x , y , and z that make all three equations true.

We can find these by doing a series of algebraic manipulations and substitutions

A classic “Word Problem.”

$$\begin{aligned}6x + 6y &= 7 \\2x + 3y + z &= 4 \\4x + 3z &= 5\end{aligned}$$

Miguel spends \$7 to buy 6 apples and 6 oranges. Liz buys 2 apples, 3 oranges and 1 mango for \$4. Wei buys 4 apples and 3 mangos, spending \$5. How much do apples, oranges and mangos cost?

$$(1) \quad 6x + 6y = 7 \quad \rightarrow \quad y = \frac{7-6x}{6} \qquad (2) \quad 4x + 3z = 5 \quad \rightarrow \quad z = \frac{5-4x}{3}$$

$$(3) \quad 2x + 3y + z = 4 \quad \rightarrow \quad 2x + 3\left(\frac{7-6x}{6}\right) + \left(\frac{5-4x}{3}\right) = 4$$

$$\rightarrow \quad \left(2 - 3 - \frac{4}{3}\right)x + \left(\frac{7}{2} + \frac{5}{3}\right) = 4$$

$$\rightarrow \quad \left(\frac{-7}{3}\right)x + \left(\frac{31}{6}\right) = 4 \quad \rightarrow \quad \left(\frac{-7}{3}\right)x = \frac{-7}{6} \quad \rightarrow \quad x = \frac{1}{2}$$

$$(4) \quad x = \frac{1}{2} \quad \rightarrow \quad y = \frac{7-6x}{6} = \frac{4}{6} = \frac{2}{3}, \quad z = \frac{5-4x}{3} = \frac{3}{3} = 1$$

A classic “Word Problem.”

Miguel spends \$7 to buy 6 apples and 6 oranges. Liz buys 2 apples, 3 oranges and 1 mango for \$4. Wei buys 4 apples and 3 mangos, spending \$5. How much do apples, oranges and mangos cost?

Let x , y , and z be the cost of apples, oranges and mangoes, respectively.

$$6x + 6y = 7$$

$$2x + 3y + z = 4$$

$$4x + 3z = 5$$

Mathematical Solution: $x = \frac{1}{2}$, $y = \frac{2}{3}$, $z = \frac{5-4x}{3} = \frac{3}{3} = 1$

Apples are 2 for \$1, oranges are 3 for \$2, and mangoes are \$1 each.

Solving problems with mathematical models.

- The execution step of even simple problems can be tedious and get a bit algebraically and/or numerically messy.
- One important application of linear algebra is the development of robust methods that can be used to solve any problem of a given type, so that this step can be implemented computationally.
- These methods can only be applied, however, once a problem is formulated into the appropriate type of mathematical model.

Unit 1.2.1

**Worked examples in formulating
mathematical models.**

Solving problems with mathematical models.

Key steps:

1. Formulate the model:
 - Carefully define the variables in the model.
 - Write one or more equations that capture all the provided detail.
2. Think about how to solve the problem.
 - What should a solution “look” like; what do you want to find?
 - What method is most appropriate to get this solution?
3. Try to solve the problem:
 - Find the mathematical solution.
 - Interpret the mathematical solution in the context of the problem.

An industrial production problem.

- An oil company has a number classes of refinery with different production outputs per barrel of crude oil:
 - Type 1 produces 15 gallons of gasoline, 5 gallons of kerosene (jet fuel), 10 gallons of diesel, 5 gallons of (heavy) fuel oil and 5 gallons of asphalt.
 - Type 2 produces 20 gallons of gasoline, 10 gallons of kerosene, and 10 gallons of diesel, with no other products.
 - Type 3 produces 10 gallons of gasoline, 5 gallons of kerosene, 5 gallons of diesel, 10 gallons of heavy fuel oil and 5 gallons of asphalt.
- How many barrels should be processed by each class in order to produce 80 M gallons of gasoline, 30 M gallons of kerosene, 40 M gallons of diesel, 45 M gallons of fuel oil and 30 M gallons of asphalt?

An industrial production problem.

- Variables are the unknown parts of the problem: the numbers of barrels of crude oil processed by each class of refinery.
- Let x_1 be the number of barrels (in millions) processed by Type 1 refineries.
- Let x_2 be the number of barrels (in millions) processed by Type 2 refineries.
- Let x_3 be the number of barrels (in millions) processed by Type 3 refineries.

An industrial production problem.

- Let x_i be the number of barrels (in millions) processed by Type “i” refineries, $i=1,2,3$.
- Each output product gives one equation, with the actual production as a function of number of barrels processed on the left-hand side and the target production values on the right:
- **Gasoline:** $15x_1 + 20x_2 + 10x_3 = 80$
- **Kerosene:** $5x_1 + 10x_2 + 5x_3 = 30$
- **Diesel:** $10x_1 + 10x_2 + 5x_3 = 40$
- **Fuel Oil:** $5x_1 + 0x_2 + 10x_3 = 45$
- **Asphalt:** $5x_1 + 0x_2 + 5x_3 = 30$

An industrial production problem.

- Let x_i be the number of barrels (in millions) processed by Type “i” refineries, $i=1,2,3$:
- Gasoline: $15x_1 + 20x_2 + 10x_3 = 80$
- Kerosene: $5x_1 + 10x_2 + 5x_3 = 30$
- Diesel: $10x_1 + 10x_2 + 5x_3 = 40$
- Fuel Oil: $5x_1 + 0x_2 + 10x_3 = 45$
- Asphalt: $5x_1 + 0x_2 + 5x_3 = 30$
- We would like to find the values of x_1 , x_2 and x_3 that simultaneously make all equations true.
- No trivial substitution plan.

An industrial production problem.

$$15x_1 + 20x_2 + 10x_3 = 80$$

$$5x_1 + 10x_2 + 5x_3 = 30$$

$$10x_1 + 10x_2 + 5x_3 = 40$$

$$5x_1 + 0x_2 + 10x_3 = 45$$

$$5x_1 + 0x_2 + 5x_3 = 30$$

- **Equation 4 – Equation 5:**

$$\begin{array}{r} (5x_1 + 0x_2 + 10x_3 = 45) \\ - (5x_1 + 0x_2 + 5x_3 = 30) \end{array} \rightarrow 5x_3 = 15 \rightarrow x_3 = 3$$

- **Equation 3 – Equation 2**

$$\begin{array}{r} (10x_1 + 10x_2 + 5x_3 = 40) \\ - (5x_1 + 10x_2 + 5x_3 = 30) \end{array} \rightarrow 5x_1 = 10 \rightarrow x_1 = 2$$

- **Equation 1**

$$15x_1 + 20x_2 + 10x_3 = 80 \rightarrow 30 + 20x_2 + 30 = 80 \rightarrow 20x_2 = 20 \rightarrow x_2 = 1$$

- **It looks like we have an answer:** $x_1 = 2, x_2 = 1, x_3 = 3$
- **But we need to verify all equations!**

An industrial production problem.

Verifying our answer: $x_1 = 2, x_2 = 1, x_3 = 3$

$$15x_1 + 20x_2 + 10x_3 = 80 \rightarrow 30 + 20 + 30 = 80 \text{ TRUE } (80=80)$$

$$5x_1 + 10x_2 + 5x_3 = 30 \rightarrow 10 + 10 + 15 = 30 \text{ FALSE } (35>30)$$

ERROR = +5

$$10x_1 + 10x_2 + 5x_3 = 40 \rightarrow 20 + 10 + 15 = 40 \text{ FALSE } (45>40)$$

ERROR = +5

$$5x_1 + 0x_2 + 10x_3 = 45 \rightarrow 10 + 0 + 30 = 45 \text{ FALSE } (40<45)$$

ERROR = -5

$$5x_1 + 0x_2 + 5x_3 = 30 \rightarrow 10 + 0 + 15 = 30 \text{ FALSE } (25<30)$$

ERROR = -5

- Our solution is inconsistent!
- This system of equations does not have an exact solution, but our production values are accurate to within 5 of the target; largest percent error is $5/30=16.7\%$.
- We may try to get an approximately solution by making some adjustments to our guess.
- Decrease x_2 a bit, and increase x_3 ?

An industrial production problem.

Guess a new solution: $x_1 = 2$, $x_2 = 0.5$, $x_3 = 3.5$

$15x_1 + 20x_2 + 10x_3 = 80$	$\rightarrow 30 + 10 + 35 = 80$	FALSE	(75<80)	ERROR = -5
$5x_1 + 10x_2 + 5x_3 = 30$	$\rightarrow 10 + 5 + 17.5 = 30$	FALSE	(32.5>30)	ERROR = +2.5
$10x_1 + 10x_2 + 5x_3 = 40$	$\rightarrow 20 + 5 + 17.5 = 40$	FALSE	(42.5>40)	ERROR = +2.5
$5x_1 + 0x_2 + 10x_3 = 45$	$\rightarrow 10 + 0 + 35 = 45$	TRUE	(45=45)	
$5x_1 + 0x_2 + 5x_3 = 30$	$\rightarrow 10 + 0 + 17.5 = 30$	FALSE	(27.5<30)	ERROR = -2.5

- This solution gives a lower maximum percentage error: $2.5/30 = 8.3\%$
- Perhaps we can do better, but it is a tedious process.
- *What is the best we can do?*

An industrial production problem.

- There is no way to produce these targets exactly!
- Processing 2 million gallons of crude with Type 1 refinery, 0.5 million gallons with Type 2, and 3.5 million gallons with Type 3 will result in production within 10% of each target.

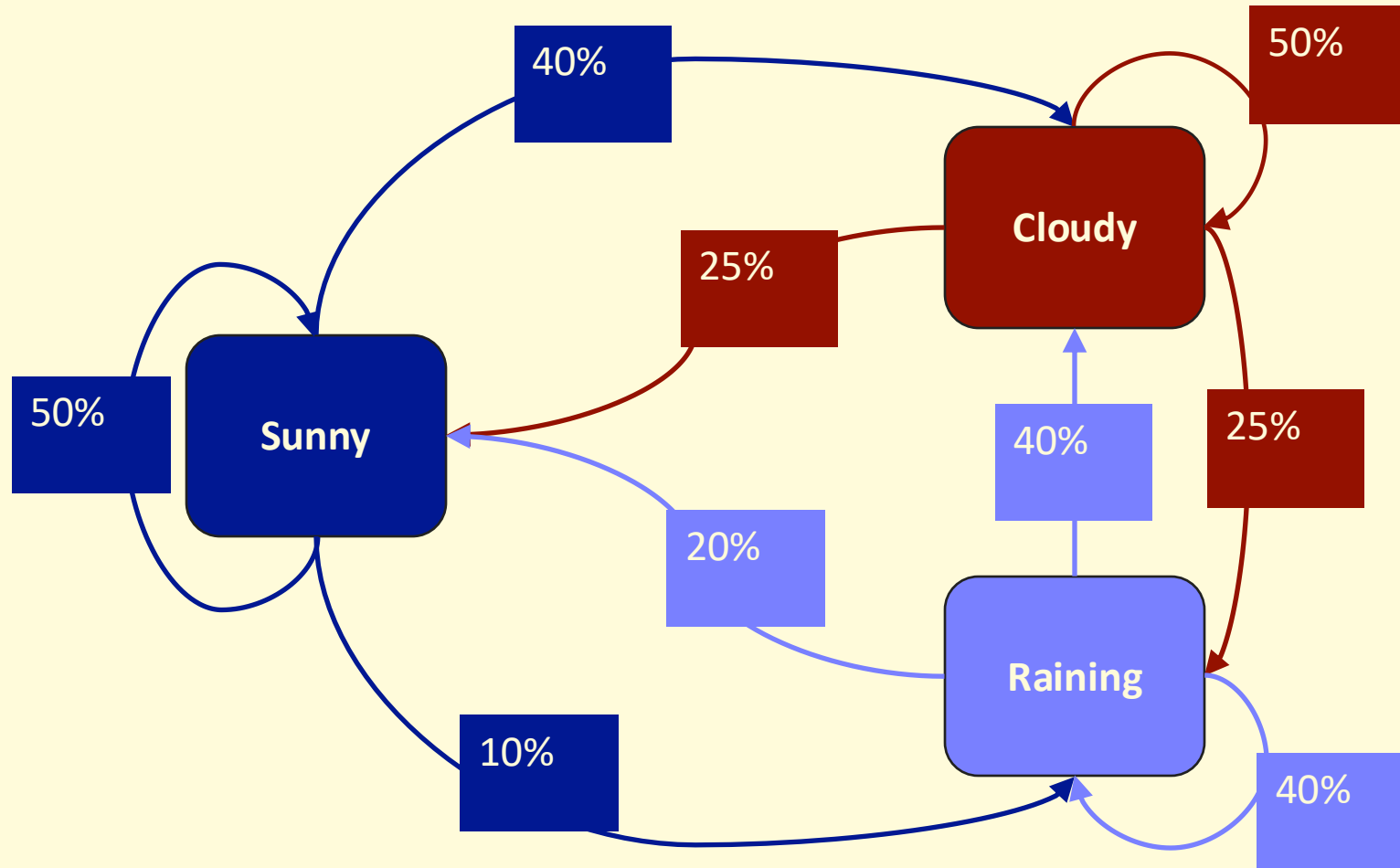
A stochastic weather prediction problem.

- If it is sunny one day, there is a 50% chance it will be sunny the next day, a 40% chance that it will be cloudy the next day, and a 10% chance that it will rain the next day.
- If it is cloudy one day, there is a 50% chance it will remain cloudy the next day, and a 25% chance each of raining or being sunny the next day.
- If it is raining one day, there is a 40% chance it will still be raining the next day, a 40% chance it will be cloudy, and a 20% chance it will be sunny.
- If it is sunny on Wednesday, what is the probability that it will be raining on Sunday?
- What is the probability it will be sunny on a randomly selected day of the year?

A stochastic weather prediction problem.

- This is a class of problem known as a Markov chain, where the key variables are the probabilities that the system (the weather in this case) is in each state at a particular point in time.
- Before writing down the equations, it can help to summarize the problem in a schematic.

A stochastic weather prediction problem.



A stochastic weather prediction problem.

		If the weather on one day is:		
		Sunny	Cloudy	Raining
The probability that the next day will be:	Sunny	50%	25%	20%
	Cloudy	40%	50%	40%
	Raining	10%	25%	40%

A stochastic weather prediction problem.

Define variables:

- Let $p_{S,i}$ be the probability of being sunny on day “i”.
- Let $p_{C,i}$ be the probability of being cloudy on day “i”.
- Let $p_{R,i}$ be the probability of being raining on day “i”.

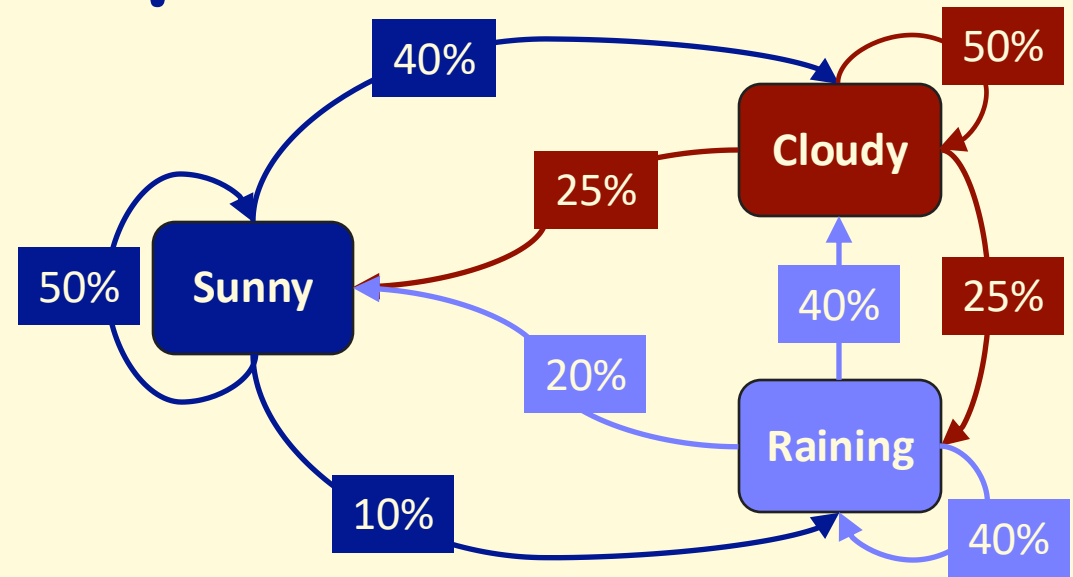
Write equations:

- One equation for each weather state.
- Variables for probabilities for day “i+1” on the left-hand side, computed from probabilities for day “i” on the right.

$$p_{S,i+1} = 0.50p_{S,i} + 0.25p_{C,i} + 0.20p_{R,i}$$

$$p_{C,i+1} = 0.40p_{S,i} + 0.50p_{C,i} + 0.40p_{R,i}$$

$$p_{R,i+1} = 0.10p_{S,i} + 0.25p_{C,i} + 0.40p_{R,i}$$



		If the weather on one day is:		
		Sunny	Cloudy	Raining
The probability that the next day will be:	Sunny	50%	25%	20%
	Cloudy	40%	50%	40%
	Raining	10%	25%	40%

A stochastic weather prediction problem.

- Requirement 1: If it is sunny on Wednesday, what is the probability that it will be raining on Sunday?
- Let $p_{X,i}$ be the probability of being in state “X” on day “i”, where X=S for sunny, X=C for cloudy and X=R for raining.

$$p_{S,i+1} = 0.50p_{S,i} + 0.25p_{C,i} + 0.20p_{R,i}$$

$$p_{C,i+1} = 0.40p_{S,i} + 0.50p_{C,i} + 0.40p_{R,i}$$

$$p_{R,i+1} = 0.10p_{S,i} + 0.25p_{C,i} + 0.40p_{R,i}$$

- Problem 1: We want to find $p_{R,4}$, given that $p_{S,0} = 1$ (and $p_{C,0} = p_{R,0} = 0$).
- An iterative solution is possible, requiring 4 steps.

A stochastic weather prediction problem.

Wed. $p_{S,0} = 1, p_{C,0} = p_{R,0} = 0$

Thu: $p_{S,1} = 0.5, p_{C,1} = 0.4, p_{R,1} = 0.1$

Fri: $p_{S,2} = 0.25 + 0.1 + 0.02 = 0.37,$

$$p_{C,2} = 0.2 + 0.2 + 0.04 = 0.44,$$

$$p_{R,2} = 0.05 + 0.1 + 0.04 = 0.19$$

Sat: $p_{S,3} = 0.185 + 0.11 + 0.038 = 0.333,$

$$p_{C,3} = 0.148 + 0.22 + 0.076 = 0.444,$$

$$p_{R,3} = 0.037 + 0.11 + 0.076 = 0.223$$

Sun: $p_{R,4} = 0.0333 + 0.111 + 0.0892 = 0.2335$

- **Note:** Given the requirement of the problem, we do not need to calculate the probability of being sunny or cloudy on day 4 (Sunday)

$$\begin{aligned} p_{S,i+1} &= 0.50p_{S,i} + 0.25p_{C,i} + 0.20p_{R,i} \\ p_{C,i+1} &= 0.40p_{S,i} + 0.50p_{C,i} + 0.40p_{R,i} \\ p_{R,i+1} &= 0.10p_{S,i} + 0.25p_{C,i} + 0.40p_{R,i} \end{aligned}$$

A stochastic weather prediction problem.

- If it is sunny on Wednesday, there is a 23.35% probability it will be raining on Sunday.

A stochastic weather prediction problem.

- Requirement 2: What is the probability it will be sunny on a randomly selected day of the year?
- Let $p_{X,i}$ be the probability of being in state “X” on day “i”, where X=S for sunny, X=C for cloudy and X=R for raining.

$$\begin{aligned}p_{S,i+1} &= 0.50p_{S,i} + 0.25p_{C,i} + 0.20p_{R,i} \\p_{C,i+1} &= 0.40p_{S,i} + 0.50p_{C,i} + 0.40p_{R,i} \\p_{R,i+1} &= 0.10p_{S,i} + 0.25p_{C,i} + 0.40p_{R,i}\end{aligned}$$

- Problem 2: One approach is to find $\lim_{i \rightarrow \infty} p_{S,i}$, given any starting probabilities (for example $p_{S,0} = 1, p_{C,0} = p_{R,0} = 0$). An iterative solution is possible but may need many, many steps.
- An alternate approach is to solve the system to have the same probabilities on days i and $i+1$; this is the “equilibrium distribution”.

A stochastic weather prediction problem.

Approach 1, a computational solution:

Day 0: $p_{S,0} = 1, p_{C,0} = p_{R,0} = 0$

Day 4: $p_{S,4} = 0.3211, p_{C,4} = 0.4444, p_{R,4} = 0.2335$

Day 5: $p_{S,5} = 0.3189, p_{C,5} = 0.4444, p_{R,5} = 0.2367$

Day 6: $p_{S,6} = 0.3179, p_{C,6} = 0.4444, p_{R,6} = 0.2377$

Day 7: $p_{S,7} = 0.3176, p_{C,7} = 0.4444, p_{R,7} = 0.2380$

Day 8: $p_{S,8} = 0.3175, p_{C,8} = 0.4444, p_{R,8} = 0.2381$

Day 9: $p_{S,9} = 0.3175, p_{C,9} = 0.4444, p_{R,9} = 0.2381$

Day 10: $p_{S,10} = 0.3175, p_{C,10} = 0.4444, p_{R,10} = 0.2381$

Day 100: $p_{S,100} = 0.3175, p_{C,100} = 0.4444, p_{R,100} = 0.2381$

- The answer does not change to within 4 significant figures after day 8: $\lim_{i \rightarrow \infty} p_{S,i} = 0.3175$

$$\begin{aligned} p_{S,i+1} &= 0.50p_{S,i} + 0.25p_{C,i} + 0.20p_{R,i} \\ p_{C,i+1} &= 0.40p_{S,i} + 0.50p_{C,i} + 0.40p_{R,i} \\ p_{R,i+1} &= 0.10p_{S,i} + 0.25p_{C,i} + 0.40p_{R,i} \end{aligned}$$

A stochastic weather prediction problem.

Approach 2:

$$\begin{aligned} p_{S,i} &= 0.50p_{S,i} + 0.25p_{C,i} + 0.20p_{R,i} & 0.50p_{S,i} - 0.25p_{C,i} - 0.20p_{R,i} &= 0 \\ p_{C,i} &= 0.40p_{S,i} + 0.50p_{C,i} + 0.40p_{R,i} & \rightarrow -0.40p_{S,i} + 0.50p_{C,i} - 0.40p_{R,i} &= 0 \\ p_{R,i} &= 0.10p_{S,i} + 0.25p_{C,i} + 0.40p_{R,i} & -0.10p_{S,i} - 0.25p_{C,i} + 0.60p_{R,i} &= 0 \end{aligned}$$

Equation 2 + 2*Equation 1: $0.6p_{S,i} - 0.8p_{R,i} = 0 \rightarrow p_{R,i} = 0.75p_{S,i}$

$$\begin{aligned} 0.50p_{S,i} - 0.25p_{C,i} - 0.15p_{S,i} &= 0 & +0.35p_{S,i} - 0.25p_{C,i} &= 0 \\ -0.40p_{S,i} + 0.50p_{C,i} - 0.30p_{S,i} &= 0 & \rightarrow -0.70p_{S,i} + 0.50p_{C,i} &= 0 \\ -0.10p_{S,i} - 0.25p_{C,i} + 0.45p_{S,i} &= 0 & +0.35p_{S,i} - 0.25p_{C,i} &= 0 \end{aligned}$$

Note that Equation 1=Equation 3=-0.5*Equation 2: $0.35p_{S,i} - 0.25p_{C,i} = 0 \rightarrow p_{C,i} = 1.4p_{S,i}$

$$p_{S,i} + p_{C,i} + p_{R,i} = 1 \rightarrow p_{S,i} + 1.4p_{S,i} + 0.75p_{S,i} = 1 \rightarrow 3.15p_{S,i} = 1 \rightarrow p_{S,i} = \frac{1}{3.15} = 0.3175$$

- The solution we get is the same as with approach 1.

A stochastic weather prediction problem.

- Let $p_{X,i}$ be the probability of being in state “X” on day “i”, where X=S for sunny, X=C for cloudy and X=R for raining.

$$p_{S,i+1} = 0.50p_{S,i} + 0.25p_{C,i} + 0.20p_{R,i}$$

$$p_{C,i+1} = 0.40p_{S,i} + 0.50p_{C,i} + 0.40p_{R,i}$$

$$p_{R,i+1} = 0.10p_{S,i} + 0.25p_{C,i} + 0.40p_{R,i}$$

- **Problem 2:** One approach is to find $\lim_{i \rightarrow \infty} p_{S,i}$, given any starting probabilities (for example $p_{S,0} = 1, p_{C,0} = p_{R,0} = 0$).
- An alternate approach is to solve the system to have the same probabilities on days i and i+1; this is the “equilibrium distribution”.
- **There is a 31.75% probability of it being sunny on a randomly selected day of the year.**

An economic supply and demand problem.

- To produce and distribute \$1M worth of energy requires \$0.1M worth of metals, \$0.3 M worth of transportation, \$0.1M worth of construction materials, and \$0.3M worth of input energy.
- To produce and distribute \$1M worth of metals requires \$0.3M of energy, \$0.2M of transportation, \$0.2M of construction materials, and \$0.1M of metals.
- To provide \$1M worth of transportation requires \$0.2M of energy, \$0.5M of metals and \$0.2M of construction materials.
- To produce and distribute \$1M of construction materials requires \$0.2M of energy, \$0.1M of construction materials, \$0.4M of transportation and \$0.1M of metals.
- If consumer demand requires \$80M of energy, \$50M of construction and \$100M of transportation, how much must each industry produce?

An economic supply and demand problem.

Variables are the unknown parts of the problem: the total supply and demand for the products of each sector of the economy, all expressed in units of millions of dollars.

- Let x_1 be the total supply of energy.
 - Let x_2 be the total supply of metals.
 - Let x_3 be the total supply of transportation.
 - Let x_4 be the total supply of construction materials.
-
- Let d_1 be the total demand for energy.
 - Let d_2 be the total demand for metals.
 - Let d_3 be the total demand for transportation.
 - Let d_4 be the total demand for construction materials.

An economic supply and demand problem.

- Let x_i be the total supply (in millions of dollars produced) by economic sector i .
- Let d_i be the total demand (in millions of dollars needed) for the output of economic sector i .
- Let the sectors be defined as: energy $i=1$; metals $i=2$; transportation $i=3$; and construction materials $i=4$.

Write model as system of equations:

- One equation for each economic sector. Total demand (as a variable) on the left-hand side; total demand as a function of total supply on the right.

$$\begin{aligned}d_1 &= 0.3x_1 + 0.3x_2 + 0.2x_3 + 0.2x_4 + 80 \\d_2 &= 0.1x_1 + 0.1x_2 + 0.5x_3 + 0.1x_4 + 0 \\d_3 &= 0.3x_1 + 0.2x_2 + 0.0x_3 + 0.4x_4 + 100 \\d_4 &= 0.1x_1 + 0.2x_2 + 0.2x_3 + 0.1x_4 + 50\end{aligned}$$

An economic supply and demand problem.

- Let x_i be the total supply (in millions of dollars produced) by economic sector i .
- Let d_i be the total demand (in millions of dollars needed) for the output of economic sector i .
- Let the sectors be defined as: energy $i=1$; metals $i=2$; transportation $i=3$; and construction materials $i=4$.

$$\begin{aligned}d_1 &= 0.3x_1 + 0.3x_2 + 0.2x_3 + 0.2x_4 + 80 \\d_2 &= 0.1x_1 + 0.1x_2 + 0.5x_3 + 0.1x_4 + 0 \\d_3 &= 0.3x_1 + 0.2x_2 + 0.0x_3 + 0.4x_4 + 100 \\d_4 &= 0.1x_1 + 0.2x_2 + 0.2x_3 + 0.1x_4 + 50\end{aligned}$$

- We would like to find a consistent solution, where the total supply is equal to the total demand for each sector of the economy.

$$d_i = x_i, \forall i$$

- A recursive approach is a possible solution.

An economic supply and demand problem.

Recursive solution:

- Guess an initial set of supplies
 - Compute the total demand.
 - Use these as the new supplies.
 - Repeat until there is no change.
-
- The initial guess should be somewhat larger than the consumer demand, but there is no fixed rule on what the guess should be.

$d_1 =$	$0.3x_1$	$+0.3x_2$	$+0.2x_3$	$+0.2x_4$	$+80$
$d_2 =$	$0.1x_1$	$+0.1x_2$	$+0.5x_3$	$+0.1x_4$	$+0$
$d_3 =$	$0.3x_1$	$+0.2x_2$	$+0.0x_3$	$+0.4x_4$	$+100$
$d_4 =$	$0.1x_1$	$+0.2x_2$	$+0.2x_3$	$+0.1x_4$	$+50$

An economic supply and demand problem.

Guess (2x the consumer demands):

- $x_1 = 160, x_2 = 0, x_3 = 200, x_4 = 100$
- $d_1 = 48 + 0 + 40 + 20 + 80 = 188$
- $d_2 = 16 + 0 + 100 + 10 + 0 = 126$
- $d_3 = 48 + 0 + 0 + 40 + 100 = 188$
- $d_4 = 16 + 0 + 40 + 10 + 50 = 116$
- Not very close, especially for sector 2: $188 > 160, 126 \gg 0, 188 < 200, 116 > 100$
- Repeat.

$d_1 =$	$0.3x_1$	$+0.3x_2$	$+0.2x_3$	$+0.2x_4$	$+80$
$d_2 =$	$0.1x_1$	$+0.1x_2$	$+0.5x_3$	$+0.1x_4$	$+0$
$d_3 =$	$0.3x_1$	$+0.2x_2$	$+0.0x_3$	$+0.4x_4$	$+100$
$d_4 =$	$0.1x_1$	$+0.2x_2$	$+0.2x_3$	$+0.1x_4$	$+50$

An economic supply and demand problem.

Guess (demands from first step):

- $x_1 = 188, x_2 = 126, x_3 = 188, x_4 = 116$
- $d_1 = 56.4 + 37.8 + 37.6 + 23.2 + 80 = 235$
- $d_2 = 18.8 + 12.6 + 94 + 11.6 + 0 = 137$
- $d_3 = 56.4 + 25.2 + 0 + 46.4 + 100 = 228$
- $d_4 = 18.8 + 25.2 + 37.6 + 11.6 + 50 = 143.2$
- **Not very close: $235 > 188, 137 > 126, 228 > 200, 143.2 > 116$**
- **Sectors 1 and 4 have errors near 25%: $(235-188)/235=25\%, (143.2-116)/116=23.5\%$**
- **Repeat.**

$d_1 =$	$0.3x_1$	$+0.3x_2$	$+0.2x_3$	$+0.2x_4$	$+80$
$d_2 =$	$0.1x_1$	$+0.1x_2$	$+0.5x_3$	$+0.1x_4$	$+0$
$d_3 =$	$0.3x_1$	$+0.2x_2$	$+0.0x_3$	$+0.4x_4$	$+100$
$d_4 =$	$0.1x_1$	$+0.2x_2$	$+0.2x_3$	$+0.1x_4$	$+50$

An economic supply and demand problem.

Guess (demands from second step):

- $x_1 = 188, x_2 = 126, x_3 = 188, x_4 = 116$
- $d_1 = 56.4 + 37.8 + 37.6 + 23.2 + 80 = 235$
- $d_2 = 18.8 + 12.6 + 94 + 11.6 + 0 = 137$
- $d_3 = 56.4 + 25.2 + 0 + 46.4 + 100 = 228$
- $d_4 = 18.8 + 25.2 + 37.6 + 11.6 + 50 = 143.2$
- Not very close: $235 > 188, 137 > 126, 228 > 200, 143.2 > 116$
- Sectors 1 and 4 have errors near 25%: $(235-188)/235=25\%, (143.2-116)/116=23.5\%$
- Repeat.

$d_1 =$	$0.3x_1$	$+0.3x_2$	$+0.2x_3$	$+0.2x_4$	$+80$
$d_2 =$	$0.1x_1$	$+0.1x_2$	$+0.5x_3$	$+0.1x_4$	$+0$
$d_3 =$	$0.3x_1$	$+0.2x_2$	$+0.0x_3$	$+0.4x_4$	$+100$
$d_4 =$	$0.1x_1$	$+0.2x_2$	$+0.2x_3$	$+0.1x_4$	$+50$

An economic supply and demand problem.

Computational solutions:

1. $x_1 = 160, x_2 = 0, x_3 = 200, x_4 = 100$
2. $x_1 = 188, x_2 = 126, x_3 = 188, x_4 = 116$
3. $x_1 = 235, x_2 = 137, x_3 = 228, x_4 = 143.2$
4. $x_1 = 265.8, x_2 = 165.5, x_3 = 255.2, x_4 = 160.8$
5. $x_1 = 292.6, x_2 = 186.8, x_3 = 277.2, x_4 = 176.8$
6. $x_1 = 314.6, x_2 = 204.2, x_3 = 295.9, x_4 = 189.7$

- **Errors now all less than 10%.**

10. $x_1 = 370.4, x_2 = 248, x_3 = 342.8, x_4 = 222.7$
20. $x_1 = 411.9, x_2 = 282.0, x_3 = 377.7, x_4 = 247.1$
50. $x_1 = 419.1, x_2 = 287.7, x_3 = 383.8, x_4 = 251.3$
100. $x_1 = 419.1, x_2 = 287.7, x_3 = 383.8, x_4 = 251.3$

- **Stopping after 6 steps gives errors up to 29% of “correct” solution. Maximum error reduced to 13.5% after 10 steps, and to 2% after 20; less than 0.01% error after 50 steps.**

$d_1 =$	$0.3x_1$	$+0.3x_2$	$+0.2x_3$	$+0.2x_4$	$+80$
$d_2 =$	$0.1x_1$	$+0.1x_2$	$+0.5x_3$	$+0.1x_4$	$+0$
$d_3 =$	$0.3x_1$	$+0.2x_2$	$+0.0x_3$	$+0.4x_4$	$+100$
$d_4 =$	$0.1x_1$	$+0.2x_2$	$+0.2x_3$	$+0.1x_4$	$+50$

An economic supply and demand problem.

- The energy industry must produce \$419.1M, the metal industry must produce \$287.7M, the transportation industry must produce \$383.8M and the construction industry must produce \$251.3M.
- This solution **CAN NOT** be obtained by hand; we require many error-prone and tedious recursive steps.

An economic supply and demand problem.

- Let x_i be the total supply (in millions of dollars produced) by economic sector i .
- Let d_i be the total demand (in millions of dollars needed) for the output of economic sector i .
- Let the sectors be defined as: energy $i=1$; metals $i=2$; transportation $i=3$; and construction materials $i=4$.

$$\begin{aligned}d_1 &= 0.3x_1 + 0.3x_2 + 0.2x_3 + 0.2x_4 + 80 \\d_2 &= 0.1x_1 + 0.1x_2 + 0.5x_3 + 0.1x_4 + 0 \\d_3 &= 0.3x_1 + 0.2x_2 + 0.0x_3 + 0.4x_4 + 100 \\d_4 &= 0.1x_1 + 0.2x_2 + 0.2x_3 + 0.1x_4 + 50\end{aligned}$$

- We would like to find a consistent solution, where the total supply is equal to the total demand for each sector of the economy.

$$d_i = x_i, \forall i$$

- **Alternative approach: Use the tools of linear algebra!**

A constrained optimization problem.

- A farmer must decide how to divide 200 acres of arable land between two major crops (wheat and corn).
- Corn is more costly to produce, requiring 4 worker-days and of labor and \$200 of capital for each acre planted;
- Wheat requires only a single worker-day of labor and \$100 of capital per acre;
- However, corn produces a profit of \$600/acre, while wheat produces a profit of only \$400/acre.
- If the farmer has \$22,000 of available capital and 320 worker-days of labor, what is the most profitable planting strategy?

A constrained optimization problem.

Variables are the unknowns in the problem: how many acres of wheat and of corn should be planted, and what is the profit.

- Let x be the number of acres of corn, y be the number of acres of wheat, and P be the total profit. We can write four expressions: one for land, one for labor, one for capital, and one for profit.
- Land: $x + y \leq 200$
- Labor: $4x + y \leq 320$
- Capital: $200x + 100y \leq 22,000$
- Profit: $P = 600x + 400y$
- We note that the first three expressions are inequalities that must be satisfied; we call these the constraints of the problem.
- The last expression allows us to compute the profit; this is the target function that we would like to maximize.

A constrained optimization problem.

Let x be the number of acres of corn, y be the number of acres of wheat, and P be the total profit.

Maximize:

$$P = 600x + 400y$$

Subject to the constraints:

$$x + y \leq 200$$

$$4x + y \leq 320$$

$$200x + 100y \leq 22,000$$

- We wish to find the values of x and y that will give us the largest possible profit, P , while satisfying all the constraints.
- We can solve this problem with graphical analysis, considering the region of “feasible solutions”.

A constrained optimization problem.

Each constraint is a line in a plot of x vs y :

- $x + y \leq 200 \quad \rightarrow y \leq 200 - x$
- $4x + y \leq 320 \quad \rightarrow y \leq 320 - 4x$
- $200x + 100y \leq 22,000 \quad \rightarrow y \leq 220 - 2x$

Maximize:

$$P = 600x + 400y$$

Subject to the constraints:

$$x + y \leq 200$$

$$4x + y \leq 320$$

$$200x + 100y \leq 22,000$$

Plot them!

Note, there are also a couple of implicit constraints: $x \geq 0$ and $y \geq 0$.

A constrained optimization problem.

Only the shaded region is allowed by the constraints (under all three lines).

Increasing x and/or y must always increase profit, so the highest profit must occur on the boundary of the feasible region.

Moving along a single line on the boundary will either increase or decrease profits (depending on direction), so the highest profit must occur at the intersection of two constraints.

Maximize:

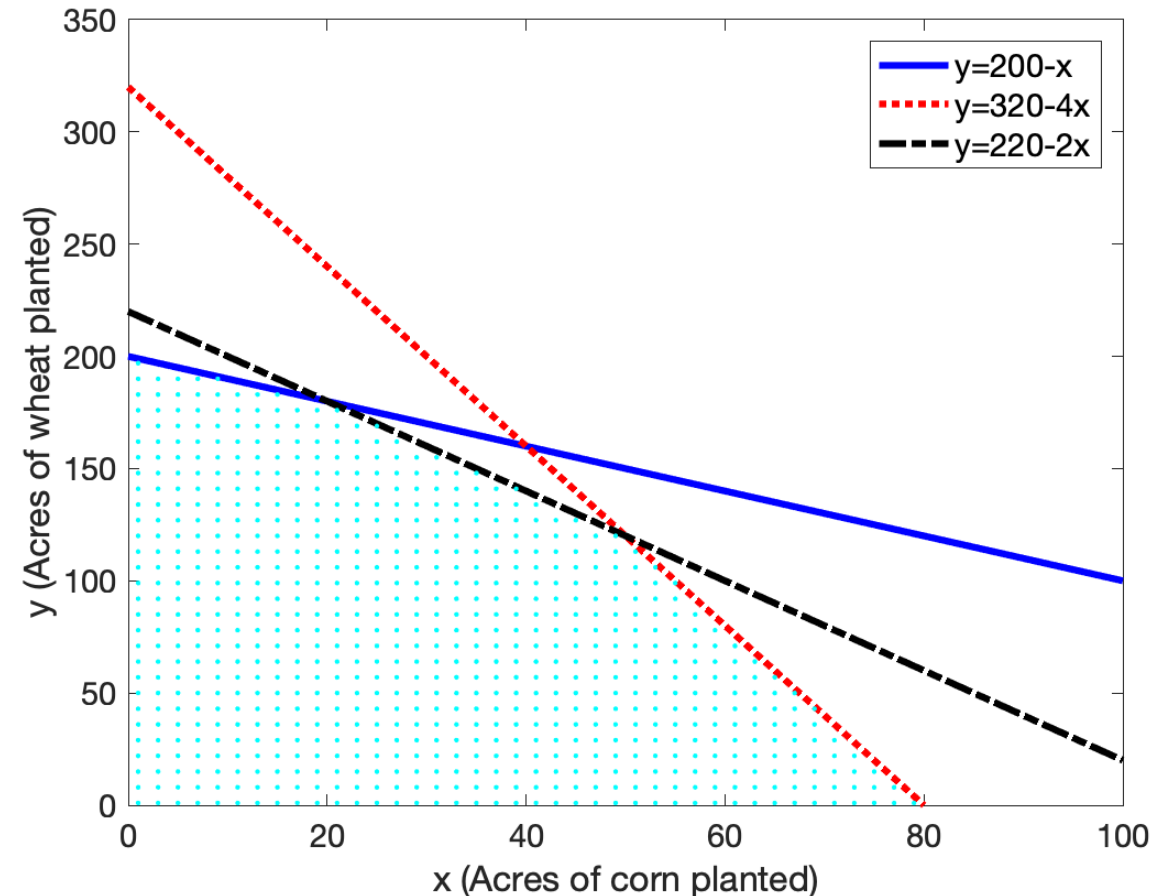
$$P = 600x + 400y$$

Subject to the constraints:

$$x + y \leq 200$$

$$4x + y \leq 320$$

$$200x + 100y \leq 22,000$$



A constrained optimization problem.

4 possible intersection points, excluding (0,0):

- $x = 0, y = 200 - x \rightarrow (x, y) = (0, 200)$
 - $y = 200 - x, y = 220 - 2x \rightarrow (x, y) = (20, 180)$
 - $y = 220 - 2x, y = 320 - 4x \rightarrow (x, y) = (50, 120)$
 - $y = 320 - 4x, y = 0 \rightarrow (x, y) = (80, 0)$
- Compute the profit at each point; the largest gives us the solution.

Maximize:

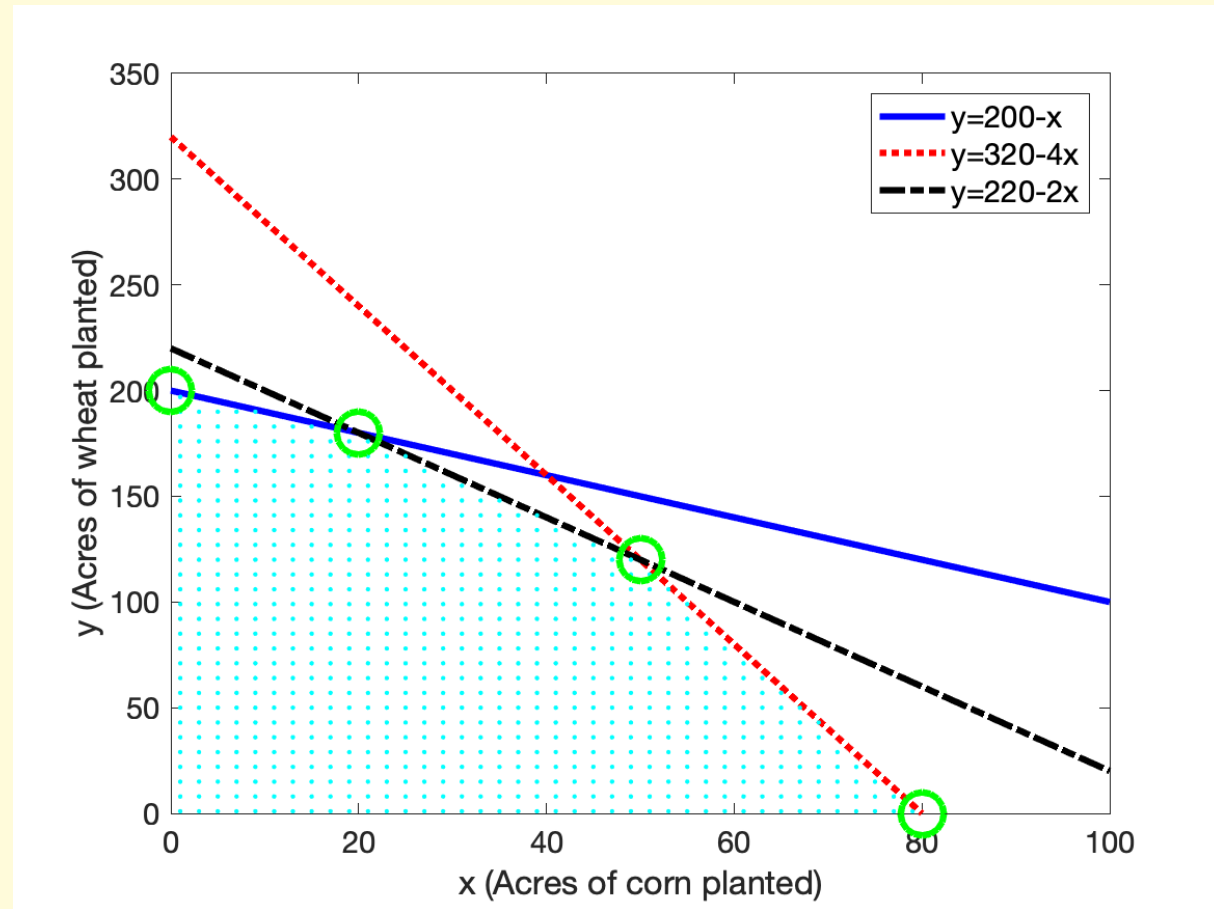
$$P = 600x + 400y$$

Subject to the constraints:

$$x + y \leq 200$$

$$4x + y \leq 320$$

$$200x + 100y \leq 22,000$$



A constrained optimization problem.

- Compute the profit at each intersection point:
- $(x, y) = (0, 200) \rightarrow P = 600(0) + 400(200) = 80,000$
- $(x, y) = (20, 180) \rightarrow P = 600(20) + 400(180) = 84,000$
- $(x, y) = (50, 120) \rightarrow P = 600(50) + 400(120) = 78,000$
- $(x, y) = (80, 0) \rightarrow P = 600(80) + 400(0) = 48,000$
- **P is maximized at $(x, y) = (20, 180)$, with a value of 84,000.**

Maximize:

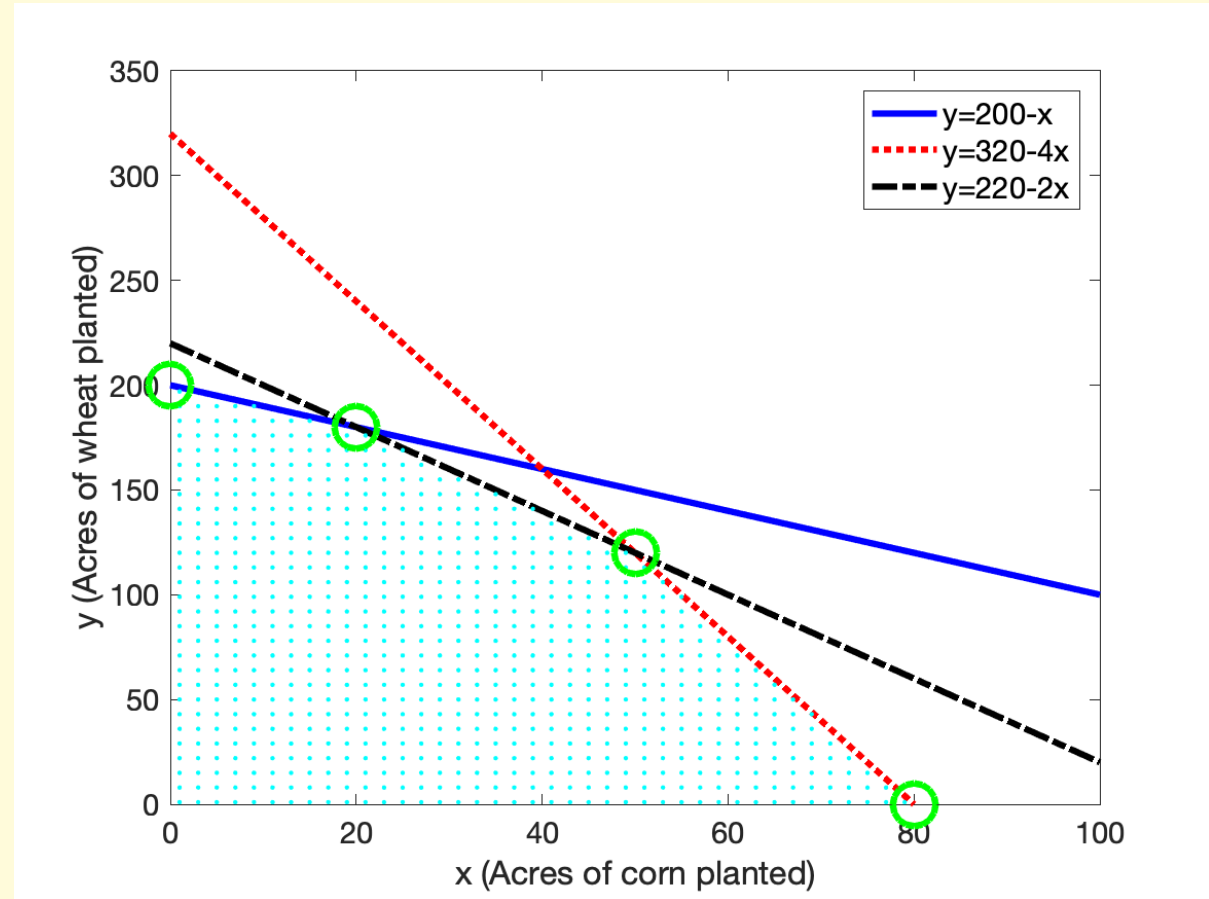
$$P = 600x + 400y$$

Subject to the constraints:

$$x + y \leq 200$$

$$4x + y \leq 320$$

$$200x + 100y \leq 22,000$$



A constrained optimization problem.

- The most profitable planting strategy is to plant 20 acres of corn and 180 acres of wheat, for a profit of \$84,000.
- We note that the limiting constraints at this point are the available land and the available capital; labor is not limiting.